Event Attribution of Frequent Pattern Dynamics: Permutation Testing with FDR Control

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Abstract—Frequent pattern mining discovers co-occurrence frequency of itemsets but lacks the ability to explain which external events (promotions, holidays, policy changes, etc.) drive temporal changes in pattern support. This paper formalizes this task as the event attribution problem and identifies its structural challenges as a multiple testing problem with FDR control: correlated false positives, temporal confounding, and hypothesis space scale. We propose a five-step pipeline comprising change point extraction, attribution scoring, circular-shift permutation testing, Benjamini–Hochberg FDR correction, and Union-Find deduplication. On synthetic data, we confirm FDR 0.070 ($\leq \alpha = 0.10$) under null conditions and demonstrate the necessity of each pipeline component through ablation analysis. Comparison against four related methods (Wilcoxon, CausalImpact, ITS, EventStudy) achieves best-or-equal F1 across all experimental conditions. Exploratory application to real-world retail data (Dunnhumby, 276K baskets) confirms 100% campaign coverage with an average of 3.0 attributions per campaign.

Index Terms—frequent pattern mining, event attribution, change point detection, permutation test, multiple testing correction, support time series

I. Introduction

Frequent pattern mining [1]–[3] is a foundational technique for discovering repeatedly co-occurring itemsets in transaction data, with broad applications in retail analytics, web mining, and bioinformatics. Classical methods such as Apriori [1], FP-Growth [2], and ECLAT [4] efficiently enumerate globally frequent patterns, while sliding-window methods [5] detect locally dense patterns within specific time intervals.

However, these methods only identify which patterns are when frequent; they cannot explain why pattern occurrence frequency fluctuates. In practice, external events such as promotional campaigns, holidays, regulatory changes, and system updates often drive the temporal variation of pattern support. Identifying "which event caused which pattern to densify, and when" would enable quantifying campaign effects, detecting side effects, and diagnosing system anomalies—insights directly relevant to decision-making.

In this work, we formalize this challenge as statistical attribution. "Attribution" here refers to identifying the statistical significance of temporal associations between support change points and external events, not causal inference. The output is positioned as a screening tool for further analysis (see Definition 5 and Remark 1).

Despite its practical importance, no statistical framework exists, to our knowledge, that links support fluctuations of frequent patterns to external events. Temporal pattern mining [6]–[9] discovers patterns with periodic or local frequency but does not model the association between pattern variation and exogenous events. Change point detection methods [10], [11] identify structural changes in individual time series but lack the capacity to handle the multiple testing burden of thousands of pattern support series simultaneously or to link detected changes to events. The econometric event study method [12] evaluates the impact of corporate events on stock prices but targets a single outcome variable and cannot be directly applied to thousands of pattern time series simultaneously.

A. Contributions

The main contributions of this work are a formal definition of the event attribution problem and, to the best of our knowledge, the first statistical solution framework. (1) We formulate the event attribution problem as a multiple testing problem with FDR control and identify three challenges: scale, secondary patterns, and temporal confounding (§III). (2) We construct a five-step pipeline comprising change point extraction, attribution scoring, circular-shift permutation testing, global BH correction, and Union-Find deduplication (§IV). (3) We identify the secondary pattern problem—where all patterns containing item i fluctuate jointly when i's frequency changes—and resolve it via deduplication based on a $\lceil l/2 \rceil$ majority-share criterion (§IV-E). (4) We provide evaluation across 8 synthetic conditions, ablation analysis, comparison against 4 related methods, and exploratory application to real-world data (Dunnhumby) (§V).

II. Related Work

A. Frequent Pattern Mining

A comprehensive survey of frequent pattern mining is provided in [3]. Apriori [1] established the anti-monotone

candidate generation and pruning framework, while FP-Growth [2] and ECLAT [4] achieved speedups via FP-tree and vertical representations, respectively. These methods target global frequency and do not model temporal variation.

B. Temporal and Periodic Pattern Mining

Segura-Delgado et al. [13] provide a comprehensive survey of temporal ARM. PFPM [7] and PPFPM [8] are periodicity mining methods targeting patterns that recur at fixed intervals, while LPFIM [6] detects frequent itemsets within time intervals using gap-based thresholds. LPPM [9] integrates locality and periodicity criteria. These methods detect when and at what cycle a pattern is frequent, but do not model the cause of variation.

C. Apriori-window: Dense Interval Mining

Apriori-window [5] differs in design philosophy from the above methods. It has no gap parameter or periodicity assumption, instead scanning a fixed-width sliding window W over a transaction sequence and directly evaluating the local support $s_P(t)$ (Definition 1) at each position t. The output is the dense interval $\mathcal{I}_P$ for each pattern P (Definition 4)—the maximal contiguous range of window positions where support continuously meets threshold θ—representing "during which period the pattern densely occurred." Candidate pattern space is efficiently pruned using Apriori's anti-monotonicity. This work adopts Apriori-window as Step 0, using its output $(\mathcal{F}, \{\mathcal{I}_P\})$ as input to the event attribution pipeline.

D. Pattern Change Monitoring and Diagnostics

Agrawal et al. [14] proposed trend detection via shape queries on rule statistics, Chakrabarti et al. [15] proposed a surprise metric based on description length, and Liu et al. [16] proposed extraction of "fundamental changes" between two time points. These methods detect and characterize changes but lack a framework for attribution to external events.

E. Change Point Detection

Change point detection methods [11] represented by the CUSUM algorithm [10] identify structural changes in individual time series, but are not equipped for the multiple testing burden of thousands of pattern support series simultaneously or for linking changes to events.

F. Event Attribution, Intervention Analysis, and Causal Inference

Interrupted time series regression (ITS) [17], the event study method [12], CausalImpact [18], and the Wilcoxon rank-sum test [19] test the effect of a known intervention on an outcome. However, these methods typically target a single or small number of outcome variables and cannot be directly applied to settings where thousands of pattern support series are simultaneously tested against multiple events.

TABLE I: Functional Comparison with Related Methods

Method	TS Change	Event Attr.	Many Patterns	Multi-test	Dedup
Temporal FPM	–	–	✓	–	–
Apriori-window	✓	–	✓	–	–
Change Point	✓	–	–	–	–
ITS/Intervention	✓	✓	–	–	–
Event Study	✓	✓	–	–	–
CausalImpact	✓	✓	–	–	–
Wilcoxon	–	✓	–	–	–
Statistical FPM	–	–	✓	✓	–
Proposed	✓	✓	✓	✓	✓

G. Statistical Pattern Mining and Multiple Testing Correction

Webb [20], [21] addressed the problem of redundant significant patterns, introducing self-sufficiency and productivity criteria to improve the relevance of discovered patterns. Terada et al. [22] minimized the Bonferroni factor via LAMP, and Llinares-López et al. [23] achieved fast significant pattern mining combining permutation testing with Tarone's testability criterion. These methods test frequency differences between two groups, whereas this work tests the association between temporal variation of support and external events. Our pipeline adopts circular-shift permutation testing [24], [25] based on the surrogate data method [26] and global BH correction [27], [28] to achieve FDR-controlled detection with higher power than Bonferroni correction [29].

H. Summary of Differences

Table I summarizes whether each method family can address the problem setting of this work. To the best of our knowledge, this work is the first integrated framework that simultaneously attributes many pattern time series to multiple events, controls multiple testing, and eliminates secondary patterns.

III. Problem Definition

A. Notation

Let $\mathcal{I} = \{i_1, i_2, \ldots, i_m\}$ be a finite universe of items and $D = [d_1, d_2, \ldots, d_N]$ a time-ordered transaction sequence. Each transaction d_t $(t = 1, \ldots, N)$ is a subset of $\mathcal{I}$. An itemset (pattern) $P \subseteq \mathcal{I}$ occurs in transaction d_t if $P \subseteq d_t$.

Definition 1 (Support Time Series). Given pattern P, window size $W \in \mathbb{Z}_{>0}$, and transaction sequence D of length N, the support time series of P is $s_P = [s_P(0), s_P(1), \ldots, s_P(N-W)]$ where

$$s_P(t) = \left|\{d_j \in D \mid t \leq j < t + W,\ P \subseteq d_j\}\right|. \quad (1)$$

Definition 2 (External Event). An external event $e = (\mathrm{id}, \mathrm{name}, t_s, t_e)$ consists of a unique identifier, a name, and a timestamp interval $[t_s, t_e]$ representing the active

period of the event. Let $\mathcal{E} = \{e_1, e_2, \ldots, e_K\}$ denote the known set of external events.

Definition 3 (Change Point (Threshold Crossing)). A position $\tau \in \{0, 1, \ldots, N - W\}$ is a change point if either:

$$s_P(\tau) \geq \theta \wedge (\tau = 0 \vee s_P(\tau - 1) < \theta), \tag{2}$$

$$s_P(\tau) < \theta \wedge \tau > 0 \wedge s_P(\tau - 1) \geq \theta. \tag{3}$$

Equation (2) is an upward crossing and (3) a downward crossing. For a dense interval $[s, e]$, the change points are s (upward) and $e + 1$ (downward, if it exists).

Definition 4 (Dense Interval). For pattern P and support threshold θ, a dense interval $[s, e] \subseteq \{0, \ldots, T - 1\}$ is a maximal contiguous interval where $s_P(t) \geq \theta$ for all $t \in [s, e]$, and either $s = 0$ or $s_P(s - 1) < \theta$, and either $e = T - 1$ or $s_P(e + 1) < \theta$. Let $\mathcal{I}_P$ denote the set of all dense intervals of P. $\mathcal{I}_P$ is provided as the output of Step 0 (Apriori-window [5]; see §IV).

B. Event Attribution Problem

Definition 5 (Event Attribution Problem). Given the set of co-occurrence patterns $\mathcal{F} = \{P \subseteq \mathcal{I} : |P| \geq 2, \text{ support} \geq \theta\}$, the dense interval sets $\{\mathcal{I}_P\}_{P \in \mathcal{F}}$, external event set $\mathcal{E}$, and significance level α, the event attribution problem is to identify all triplets $(P, I, e) \in \{(P, I, e) : P \in \mathcal{F},\ I \in \mathcal{I}_P,\ e \in \mathcal{E}\}$ where the change points of dense interval $I \in \mathcal{I}_P$ are statistically significantly associated with the time window of event e, while controlling the false discovery rate (FDR) at level α. Singleton patterns ($|P| = 1$) are excluded, as their frequency variation is captured by standard time series analysis and lies outside the scope of co-occurrence discovery.

Remark 1 (Scope and Limitations of Attribution). "Attribution" in this paper refers to the significance of statistical temporal association, not causal inference (do-calculus [30]). When confounders exist or multiple event time windows overlap, the output of this method is positioned as a screening tool for attribution candidates (experimentally confirmed as accuracy degradation under CONFOUND and OVERLAP conditions; see §V-B).

The three main challenges of this problem are:

1) Scale: $\sum_{P \in \mathcal{F}} |\mathcal{I}_P| \times |\mathcal{E}|$ (P, I, e) hypotheses must be tested simultaneously, requiring rigorous multiple testing correction.
2) Secondary patterns: When the frequency of item i increases, the support of all co-occurrence patterns containing i fluctuates jointly, generating a large number of false positives.
3) Temporal confounding: When the time windows of multiple events overlap, attribution becomes ambiguous.

IV. Proposed Method

This section describes the event attribution pipeline in detail.As Step 0, Apriori-window [5] takes a transaction sequence D, window size W, and minimum support threshold θ as input and outputs a pattern set $\mathcal{F}$ and the dense interval sets $\{\mathcal{I}_P\}_{P \in \mathcal{F}}$. Steps 1–5 then statistically identify significant pattern–interval–event attributions. Figure 1 shows the pipeline overview; Algorithm 1 shows the attribution scoring and statistical testing procedure; Algorithm 2 shows the deduplication procedure.

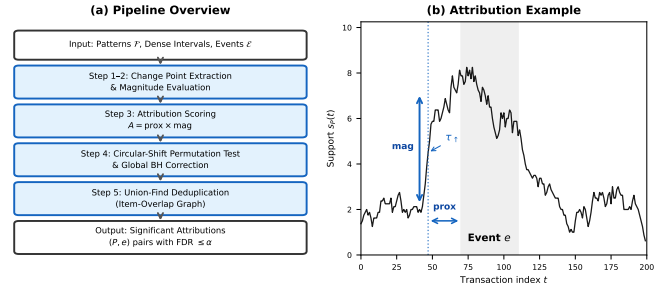

Fig. 1: (a) Overall pipeline structure. Step 0 (Apriori-window) outputs dense intervals; Steps 1–5 yield significant attributions. (b) Attribution scoring example. Proximity (prox) and magnitude (mag) are computed from the relationship between change points $\tau_\uparrow, \tau_\downarrow$ on the support time series and the event interval.

A. Step 1: Change Point Extraction from Dense Intervals

Each dense interval is treated as an independent attribution unit; if a pattern has multiple dense intervals, each can be attributed to a different event. From each dense interval $[s_i, e_i]$, we extract the upward change point s_i and downward change point $e_i + 1$ (if it exists), and compute the local support needed for the magnitude (Eq. (4)) via binary search on the pattern occurrence list T_P in $O(\log |T_P|)$.

B. Step 2: Magnitude Evaluation

Let $T = N - W + 1$ and define the reference windows of h points before and after change point τ:

$$\mathcal{L}_\tau^- = \{\max(0, \tau - h), \ldots, \tau - 1\},$$
$$\mathcal{L}_\tau^+ = \{\tau, \ldots, \min(T - 1, \tau + h - 1)\}$$

(default $h = 20$). The magnitude is defined as the local mean difference:

$$\bar{s}^\pm(\tau) = \frac{1}{\max(1, |\mathcal{L}_\tau^\pm|)} \sum_{j \in \mathcal{L}_\tau^\pm} s_P(j),$$
$$\text{magnitude}(\tau) = |\bar{s}^+(\tau) - \bar{s}^-(\tau)|. \tag{4}$$

The detection position is determined by threshold crossing (endpoints of the dense interval), and the change strength is evaluated by the level difference before and after the crossing.

Algorithm 1 Attribution Scoring and Statistical Testing (Steps 1–4)

Require: Co-occurrence patterns $\mathcal{F}$ ($|P| \geq 2$), dense intervals $\{\mathcal{I}_P\}$, events $\mathcal{E}$, settings $(\theta, \sigma, B, \alpha)$
Ensure: Significant attribution set $\mathcal{R}$

```
// Steps 1–2: Change point extraction and magnitude computation
RawResults ← []
for each pattern P ∈ F, each interval [s_i, e_i] ∈ I_P do
    C_I ← {s_i (upward), e_i+1 (downward)}
    Compute magnitude(τ) for each τ ∈ C_I   ▷ Eq. (4)
// Step 3: Attribution scoring
    for each event e ∈ E do
        A_obs ← Σ_{τ∈C_I} A(P, τ, e)   ▷ Eq. (5)
// Step 4: Circular shift permutation test
        count ← 0
        for b = 1 to B do
            E_b ← CircularShift(E, δ_b)
            e_b ← shifted version of event e in E_b   ▷ same identifier as e
            A_b ← Σ_{τ∈C_I} A(P, τ, e_b)
            if A_b ≥ A_obs then count ← count + 1
            end if
        end for
        p_raw ← (1 + count)/(B + 1)   ▷ Eq. (9)
        RawResults.append((P, [s_i, e_i], e), A_obs, p_raw)
    end for
end for
// Step 4 (cont.): Global BH correction
Sort RawResults by p_raw in ascending order
M ← |RawResults|
for i = M downto 1 do
    p̃_(i) ← min(p̃_(i+1), min(1, p_(i) · M/i))   ▷ BH step-down
end for
R ← {(P, I, e) | p̃_(i) < α}
return R
```

C. Step 3: Attribution Scoring

For each change point τ of dense interval $I = [s_i, e_i]$ and each event $e \in \mathcal{E}$, we compute an attribution score that quantifies temporal association. We adopt:

$$A(P, \tau, e) = \text{prox}(\tau, e) \cdot \text{mag}(\tau) \tag{5}$$

The two components are as follows.

Proximity: A measure of temporal closeness via exponential kernel:

$$\text{prox}(\tau, e) = \exp\left(-\frac{d(\tau, e)}{\sigma}\right) \tag{6}$$

where $d(\tau, e) = \min(|\tau - t_s|, |\tau - t_e|)$ is the minimum distance from the change point to the event boundary, and $\sigma > 0$ is a decay parameter.

Magnitude: The raw magnitude from Eq. (4) (absolute level shift) normalized by square root for scoring:

$$\text{mag}(\tau) = \frac{\text{magnitude}(\tau)}{\sqrt{\max(1,\ \bar{s}^-(\tau))}} \tag{7}$$

Dividing by the square root mitigates the bias where patterns with higher baseline support tend to have larger absolute changes. Item frequencies in real-world transaction data often follow a Zipf distribution [3], making this normalization particularly effective.

The observed score for event e at dense interval I is the sum of attribution scores over the change points of that interval (at most two: upward and downward): $A_{\text{obs}}(P, I, e) = \sum_{\tau \in \mathcal{C}_I} A(P, \tau, e)$.

a) Design Rationale: The attribution score A is an index for ranking candidates; the final attribution decision is made by the permutation test in Step 4. The multiplicative structure prox × mag functions as an exclusive filter: if either component is near zero, the overall score approaches zero. EX2 (§V-C) empirically demonstrates the complementarity of both components.

D. Step 4: Permutation Testing and Global FDR Correction

1) Circular Shift Null Distribution: To test whether the temporal association between change points of dense interval I and event e is statistically significant, we use a circular shift permutation test. Under the null hypothesis H_0: "the positions of the change points in dense interval I are unrelated to the event time", B permutations are performed. In each permutation b, all event times are shifted simultaneously by a random offset $\delta_b \sim \text{Uniform}\{1, \ldots, T-1\}$ (circular shift):

$$e_b = (\text{id}, \text{name}, (t_s + \delta_b) \bmod T, (t_s + \delta_b + \Delta) \bmod T) \tag{8}$$

where $\Delta = t_e - t_s$ is the event duration and T is the time series length.

The score $A^{(b)}_{\text{perm}}(P, I, e)$ is recomputed for each permutation, and the uncorrected p-value is:

$$p_{\text{raw}}(P, I, e) = \frac{1 + |\{b : A^{(b)}_{\text{perm}}(P, I, e) \geq A_{\text{obs}}(P, I, e)\}|}{B + 1} \tag{9}$$

Proposition 1 (Validity of Permutation Test). Under null hypothesis H_0, the observed score A_{obs} and B permuted scores $A^{(1)}_{\text{perm}}, \ldots, A^{(B)}_{\text{perm}}$ are exchangeable. Therefore, the p-value in Eq. (9) follows a discrete uniform distribution on $\{1/(B+1), 2/(B+1), \ldots, 1\}$, and the Type I error rate is controlled at α for any level α [24].

2) Global Benjamini–Hochberg Correction: The Benjamini–Hochberg (BH) procedure [27] is applied globally to all $M = \sum_{P \in \mathcal{F}'} |\mathcal{I}_P| \cdot |\mathcal{E}|$ hypotheses ((P, I, e) triplets), maintaining FDR at level α while achieving higher power than Bonferroni correction.

See Appendix A for the PRDS condition justification.

Algorithm 2 Item Deduplication (Step 5)

Require: Significant attribution set $\mathcal{R}$ (output of Algorithm 1)
Ensure: Deduplicated attribution set $\mathcal{R}'$

```
// Step 5a: Same-length pattern deduplication (majority-share)
for each event e do
    R_e ← all patterns attributed to event e (across intervals), grouped by length l
    for each length-l group do
        for each pattern pair (P_i, P_j) do
            if |P_i ∩ P_j| ≥ ⌈l/2⌉ then
                Union(P_i, P_j)                ▷ Union-Find
            end if
        end for
        Within each connected component, retain only the pattern with maximum A_obs(P, I, e)
    end for
end for
// Step 5b: Cross-length subset elimination
for each event e do
    for each pattern pair (P_i, P_j) where |P_i| ≠ |P_j| do
        if P_i ⊆ P_j or P_j ⊆ P_i then
            Union(P_i, P_j)                    ▷ Union-Find
        end if
    end for
    Within each connected component, retain only the pattern with maximum A_obs(P, I, e)
end for
return R' ← set of retained patterns
```

E. Step 5: Item Deduplication

1) The Secondary Pattern Problem: When an external event increases the frequency of item i, the support of all co-occurrence patterns containing i fluctuates jointly, causing secondary patterns such as $(5, 42)$, $(15, 78)$ to also pass the permutation test. Since secondary patterns genuinely exhibit support variation, statistical testing alone cannot eliminate them.

2) Deduplication via Union-Find Clustering: We resolve this issue with the following post-processing steps:

1) For each event e, collect all patterns significantly attributed to e (consolidated across intervals) and group by pattern length l.
2) Within each length-l group, construct a majority-share graph: nodes are patterns, with an edge between patterns if $|P_i \cap P_j| \geq \lceil l/2 \rceil$.
3) Find connected components using a Union-Find (disjoint set) data structure.
4) Within each connected component, retain only the pattern with the highest attribution score $A_{\text{obs}}(P, I, e)$.
5) Cross-length subset elimination: After the same-length deduplication above, apply Union-Find over all attributed patterns for event e, merging pairs where $P_i \subseteq P_j$, and retain only the highest-scoring pattern per component.

Claim 1 (Validity of Deduplication (Generalized)). Assume the simplified model of independent occurrence and additive boost. For true pattern P^* (length l, all items boosted with probability $\beta > 0$) and secondary pattern P' sharing s items with P^* (remaining $l-s$ items unboosted), $\text{magnitude}(\tau, P^*) > \text{magnitude}(\tau, P')$ holds. Furthermore, when $s \geq \lceil l/2 \rceil$, both belong to the same connected component in the majority-share graph, and P^* with the highest score is retained while P' is eliminated.

The proof is given in Appendix C.

See Appendix B for complexity analysis.

V. Experiments

A. Experimental Design

1) Synthetic Data Generation: We generate synthetic data with a base vocabulary of 200 items (per-item occurrence probability $p_{\text{base}} = 0.03$) and $N = 100{,}000$ transactions. External event effects are modeled as planted signals. Three signal types are mixed in a single dataset to multi-dimensionally evaluate the pipeline's discrimination ability:

Type A (vocab-in boost): During the event active period, all pattern items are simultaneously inserted with probability β, raising support. A realistic condition in which baseline support exists at p_{base} outside the event period.

Type B (event-unrelated dense patterns): Support rises in a specific period, but not corresponding to any event (simulates seasonality or trend).

Type C (decoy events): Events unrelated to any pattern variation.

2) Evaluation Metrics: Evaluation is performed at the (P, I, E) triplet level.

a) Ground Truth Construction: For each planted signal (P, E), the longest dense interval overlapping the event window is registered as the representative I^*, yielding ground truth triplet (P, I^*, E) (one per (P, E) pair).

b) Matching Criterion: Exact match on item set, interval endpoints, and event ID is required.

c) Precision Metrics: Standard Precision, Recall, and F1 score based on $\text{TP} = |\hat{\mathcal{A}} \cap \mathcal{A}^*|$, $\text{FP} = |\hat{\mathcal{A}} \setminus \mathcal{A}^*|$, $\text{FN} = |\mathcal{A}^* \setminus \hat{\mathcal{A}}|$. Evaluation is performed on the final output after deduplication (Step 5). Type B (event-unrelated) patterns output as significant attributions count as FP and reduce Precision.

3) Default Settings: Default parameters are shown in Appendix I (Table V). Square root normalization (Eq. (7)) is used for magnitude normalization.

4) Evaluation Policy: EX1 evaluates attribution accuracy under diverse data conditions. EX2 demonstrates the contribution of each step through ablation analysis. EX3 performs a step-difference comparison replacing Steps 1, 3, and 4 with related methods. EX4 confirms real-world

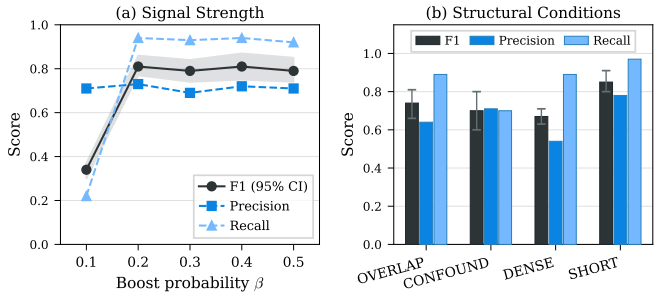

Fig. 2: EX1 results (20-seed mean with 95% CI). (a) Signal strength: $\beta \geq 0.2$ achieves F1 ≥ 0.78 and Recall ≥ 0.92. $\beta = 0.1$ reaches the detection limit for $L = 2$ vocab-in patterns (F1 $= 0.34$), while $L = 3, 4$ vocab-out patterns are stably detected. (b) Structural conditions: F1 ≥ 0.67 across all conditions (SHORT: 0.85, OVERLAP: 0.74, CONFOUND: 0.70, DENSE: 0.67).

data applicability through exploratory analysis; quantitative evaluation is not performed because ground truth attribution labels are unavailable for real-world data.

B. EX1: Core Attribution Accuracy

RQ: Can the pipeline correctly attribute event-driven variation and reject event-unrelated dense patterns? Can it naturally handle multiple pattern lengths $L = 2$–4? We evaluate robustness across diverse data characteristics (signal strength, structural conditions, pattern length).

a) Signal Strength and Structural Conditions: Type A (5 vocab-in boost patterns: $L = 2 \times 3$, $L = 3 \times 1$, $L = 4 \times 1$), Type B (2 event-unrelated patterns), and Type C (2 decoys) are mixed, evaluated with 20 seeds across $\beta \in \{0.1, 0.2, 0.3, 0.4, 0.5\}$ and 4 structural conditions (Fig. 2).

At $\beta = 0.1$, $L = 2$ vocab-in patterns reach the detection limit while $L = 3, 4$ vocab-out patterns are stably detected: F1 $= 0.34$ [0.30, 0.37] (Recall $= 0.22$). At $\beta \geq 0.2$, Recall ≥ 0.92 and F1 ≥ 0.78 are achieved. F1 ≥ 0.67 is achieved across all structural conditions (see §H).

b) Null Condition Validation: Running 100 seeds on null data with no planted signals (Appendix F), the mean per-seed FDR was 0.070 ($\leq \alpha = 0.10$), empirically confirming FDR control.

C. EX2: Ablation Analysis

RQ: How does each score component and pipeline step contribute to accuracy? We use Zipf-distributed data ($\sqrt{}$ normalization) as in EX1.

a) Score Components: We design two scenarios demonstrating the necessity of each component and apply all 3 ablation modes to each scenario (2 scenarios $\times$ 3 modes $\times$ 5 seeds $=$ 30 conditions):

Scenario A (prox required): Pattern $(5, 15)$ has two dense intervals—one near the event (causal) and one distant (accidental). Without prox, both receive equal scores.

TABLE II: EX2a Results: Per-Scenario Ablation (5-seed mean F1, Zipf-1.0, $\sqrt{}$ normalization)

Configuration	Scenario A	Scenario B
Full: prox $\times$ mag	1.00	0.46
mag only (prox removed)	0.00	0.00
prox only (mag removed)	1.00	0.19

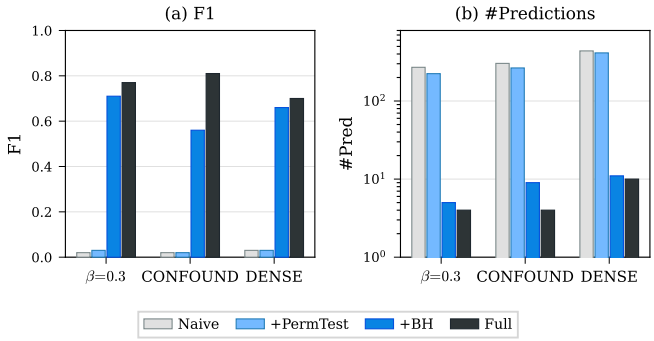

Fig. 3: EX2b: Accuracy change from incremental step additions (5-seed mean, Zipf-1.0, $\sqrt{}$ normalization). (a) F1: substantial improvement from BH correction and deduplication. (b) #Predictions: each step progressively reduces output volume, suppressing false positives.

Scenario B (mag required): Pattern $(5, 15)$ is affected by 2 events with different boost magnitudes ($\beta = 0.5$ vs. $\beta = 0.1$). Without mag, both are treated equally.

Results are shown in Table II.

Without prox (mag only), the method has no temporal alignment information, collapsing to F1 $= 0.00$ in both scenarios. Prox only achieves F1 $= 1.00$ on Scenario A but drops to F1 $= 0.19$ on Scenario B (distinguishing strong vs. weak boost). The full configuration achieves 1.00 on Scenario A and 0.46 on Scenario B, demonstrating the complementarity of both components.

b) Pipeline Step Contributions: We compare a naive baseline (Steps 1–3 only: change point detection + scoring, outputting all candidates above threshold) against three variants obtained by incrementally adding pipeline steps:

- Naive: Steps 1–3 only (no testing)
- +PermTest: Per-pattern permutation test added (no global correction)
- +BH: Global BH correction added
- Full: Union-Find deduplication added (= complete pipeline)

Using EX1 data ($\beta = 0.3$, CONFOUND, DENSE; Zipf-1.0, $\sqrt{}$ normalization), evaluated with 5 seeds per condition (Fig. 3).

Naive achieves Recall $= 1.00$ across all conditions but F1 ≤ 0.03 due to massive false positives—score thresholding alone is insufficient. +PermTest provides limited FP reduction, but +BH (global BH correction) delivers the largest false positive reduction, and Full (+deduplication) further eliminates redundant patterns

Fig. 4: EX3 results: comparison with related methods (5-seed mean F1). All methods share Step 0 (Apriori-window), BH correction, and deduplication. The proposed method achieves the highest or equal F1 across all 5 conditions, with the most prominent advantage on CONFOUND (0.68 vs. runner-up ITS 0.45).

to achieve the best F1 across all conditions. Each step is confirmed to reduce false positives in a progressive and complementary manner.

D. EX3: Comparison with Related Methods

RQ: Is the proposed component design optimal when the pipeline's Steps 1–3–4 are replaced with related methods? Comparison Purpose and Design Step 0 (Apriori-window), BH correction, and Union-Find deduplication (Step 5) are shared by all methods; only the attribution core (Steps 1, 3, 4) is replaced. Results should be interpreted as "proposed step design vs. alternative step designs."

From the related methods in Table I, we select methods that test the association between events and outcomes, replacing Steps 1 (change point extraction), 3 (attribution scoring), and 4 (statistical testing) with the following four methods:

1) Wilcoxon [19]: Rank-sum test comparing in-event vs. out-of-event support distributions.
2) CausalImpact [18]: Local-level model estimates counterfactual and tests cumulative deviation.
3) ITS [17]: Segmented regression tests level change coefficient before and after intervention.
4) EventStudy [12]: Cumulative abnormal support tested via z-test from pre-event statistics.

Results on the same data as EX1 ($\beta = 0.3$, Zipf-1.0, $\sqrt{}$ normalization) and 4 structural conditions (5 seeds each) are shown in Fig. 4.

The proposed method achieves the highest or equal F1 across all 5 conditions (near-tied with ITS on SHORT: 0.72 vs. 0.77), particularly exceeding runner-up ITS on CONFOUND (F1 = 0.68 vs. 0.45). The proximity score leveraging change point timing effectively discriminates event-unrelated patterns whose active periods overlap with event windows.

ITS is competitive at $\beta = 0.3$ (F1 = 0.60 vs. proposed: 0.66) but degrades significantly on CONFOUND (0.45) and OVERLAP (0.40). CausalImpact and Wilcoxon remain at F1 $\leq$ 0.49 across all conditions. The proposed method achieves F1 0.66–0.72 (5-condition average 0.68), with consistent stability and marked advantage under confounding and overlap conditions.

TABLE III: EX4 Results: Dunnhumby Large-Scale Promotional Campaigns

Metric	Value
Transactions	276,484
Patterns ($\|P\| \geq 2$)	896
Campaigns	5
Significant attributions (post-dedup)	15
Mean attributions per campaign	3.0

E. EX4: Real-World Data Validation

RQ: Is the method applicable to real-world data?

Exploratory analysis on 276,484 purchase baskets from the Dunnhumby "The Complete Journey" dataset [31]. Square root normalization is used, as item frequencies follow a Zipf-like distribution.

a) External Event Selection: This pipeline targets population-level support variation (Definition 5). The Dunnhumby dataset records 30 promotional campaigns, but targeted campaigns (TypeB/C: fewer than 2,000 coupon items) target specific households with individual coupons and are not expected to produce detectable population-level effects. We therefore use the 5 large-scale promotional campaigns (C8, C13, C18, C26, C30; 3.7K–38.2K coupon items) with at least 3,000 coupon items as external events. This selection is based on campaign metadata (coupon.csv) as a pre-hoc filter, not selected post-hoc by consulting attribution results.

Settings: $W = 300$, $\theta = 5$, $B = 5{,}000$, $\alpha = 0.10$ (BH correction), deduplication enabled.

At least one significant attribution was generated for all 5 campaigns (100% coverage), with an average of 3.0 attributions per campaign—a manageable volume for analysts. Quantitative accuracy evaluation is not performed because ground truth labels are unavailable, but the product categories of attributed patterns qualitatively align with campaign content, confirming applicability to real-world data.

VI. Conclusion

This paper formally defined the event attribution problem and constructed a five-step attribution pipeline as the first statistical solution.

Experiments yielded the following findings:

- High attribution accuracy was achieved across diverse data conditions including Zipf distributions; under null conditions over 100 seeds, a mean FDR of 0.070

($\leq \alpha = 0.10$) was confirmed, demonstrating FDR control (EX1).

- Ablation analysis demonstrated the complementarity of proximity and magnitude, and the progressive false positive reduction effect of global BH correction and deduplication (EX2).
- Comparison against 4 related methods confirmed that the proposed method achieves the best or equal F1 across all 5 conditions (mean F1 = 0.68), with particularly prominent advantage under confounding and overlap conditions (EX3).
- Exploratory application to real-world data (Dunnhumby, 276K baskets) confirmed attribution generation for all campaigns (100% coverage), demonstrating applicability (EX4).

A. Limitations and Future Work

This pipeline has the following limitations. First, accuracy degrades under conditions of overlapping or high-density event windows. Second, the circular shift permutation test assumes weak stationarity; data with strong trends or seasonality may require extensions such as block permutation. Third, there is a tradeoff between detection power and weak signal recovery. Future work includes handling non-stationary time series and evaluating external validity on large-scale real-world data.

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Appendix A
PRDS Condition Justification

For the BH procedure to control FDR for non-independent test statistics, the p-values must satisfy the positive regression dependency on a subset (PRDS) condition [28]. We do not provide a formal proof here, but the positive correlation between support time series of patterns sharing items induces positive dependence among corresponding p-values, serving as the practical rationale for applying PRDS.

Appendix B
Computational Complexity

The dominant bottleneck is the permutation testing in Steps 3–4, with overall complexity $O(\sum_P |\mathcal{I}_P| \cdot |\mathcal{E}| \cdot B)$. Rayon parallelization at the pattern–interval level substantially reduces practical processing time. The deduplication step has complexity $O(n \cdot k)$ per event (n: significant attributions, k: max pattern length), negligible compared to permutation testing.

Appendix C
Proof of Deduplication Validity

Proof of Claim 1. Under the independent occurrence assumption, the expected support of P^* under the influence of event e is $(p_{\text{base}} + \beta)^l$, and p_{base}^l outside the event period. Setting $r = (p_{\text{base}} + \beta)/p_{\text{base}} > 1$, we have $\text{magnitude}(P^*) \propto p_{\text{base}}^l(r^l - 1)$. Secondary pattern P' has s boosted items and $l - s$ unboosted items, so $\text{magnitude}(P') \propto p_{\text{base}}^l(r^s - 1)$. Since $r > 1$ and $l > s$, we have $r^l > r^s$, thus $\text{magnitude}(P^*) > \text{magnitude}(P')$ holds for any pattern length l. Note that this result depends on the independent occurrence assumption; when items have strong correlations, the ordering may reverse. □

Appendix D
Synthetic Data Generation Details

A. Default Conditions

The base vocabulary contains 200 items ($|\mathcal{I}| = 200$), each with occurrence probability $p_{\text{base}} = 0.03$, and $N = 100{,}000$ transactions. The expected number of items per transaction is $200 \times 0.03 = 6.0$.

- Type A (boost, 5 patterns, mixed lengths): $L = 2$ vocab-in patterns ($(5, 15)$, $(25, 35)$, $(45, 55)$) have baseline support at p_{base} (in-vocabulary). The $L = 3$ pattern ($(201, 202, 203)$) and $L = 4$ pattern ($(205, 206, 207, 208)$) use out-of-vocabulary item IDs (> 200) with zero background probability. Each pattern's event duration is 6,000 transactions (6% of total), equally spaced on the time axis. During the event period, each item is additionally inserted with probability β.
- Type B (event-unrelated, 2 patterns): Patterns $(65, 75)$ and $(85, 95)$ receive a boost of β in a random 6,000-transaction interval, but do not correspond to any event.
- Type C (decoy events, 2): Events unrelated to any pattern variation (duration 6,000), randomly placed.

B. Structural Conditions

- OVERLAP: Events E1= $[16{,}000, 28{,}000]$ and E2= $[24{,}000, 36{,}000]$ overlap in $[24{,}000, 28{,}000]$. Tests temporal discrimination ability.
- CONFOUND: The active interval of Type B patterns is intentionally placed at event time windows (confounding condition).
- DENSE: Doubled to Type A 6 patterns + Type B 4 patterns + 4 decoys (high-density condition).
- SHORT: Event duration shortened to 1,600 (approximately 27% of the default 6,000; short event condition).

C. EX2 Ablation Scenarios

- Scenario A (prox required): Pattern $(5, 15)$ has two dense intervals— one near the event at $[15{,}000, 25{,}000]$ (causal), and one at $[60{,}000, 68{,}000]$ (accidental). Event interval: $[15{,}000, 25{,}000]$.
- Scenario B (mag required): Pattern $(5, 15)$ is affected by two events— E1 with boost $\beta = 0.5$ (strong) and E2 with $\beta = 0.1$ (weak).

D. Zipf Distribution Parameters

Occurrence probability of item k ($k = 1, \ldots, 200$): $p(k) = \min(C/k^{\alpha_z},\ 0.10)$. C is set so that the median item ($k = 100$) has probability 0.03, with the maximum probability capped at 0.10.

Appendix E
Dunnhumby Dataset Details

An overview of the Dunnhumby "The Complete Journey" dataset [31] is given in Table IV.

TABLE IV: Dunnhumby Dataset Statistics

Metric	Value
Households	2,500
Transactions	276,484
Unique products	92,339
Mean items per transaction	9.39
Observation period	711 days
Campaigns	30

The 30 campaigns are classified into three types. Based on the problem definition (Definition 5) targeting population-level support variation, this experiment uses only the 5 TypeA campaigns as external events (see §V-E):

- TypeA (5 campaigns: C8, C13, C18, C26, C30): Large-scale promotions. 3,700–38,200 coupon items, 40–56 day duration.
- TypeB (18 campaigns): Targeted campaigns. 200–2,000 coupon items, 32–61 day duration.
- TypeC (7 campaigns): Niche campaigns. 200–1,500 coupon items, 32–162 day duration.

Pipeline settings differ from Table V: $W = 300$ and $\theta = 5$. W was chosen as an appropriate temporal resolution for 276K transactions.

Appendix F
Null FDR Verification Details (100 Seeds)

We verified FDR control by the BH procedure under null conditions over 100 seeds.

a) Settings: No planted signals (0 Type A patterns), 5 randomly placed decoy events (each lasting 6,000 transactions), 4 Type B dense patterns (temporally unrelated to any event), $N = 100{,}000$, $W = 1{,}000$, $\theta = 100$, $B = 5{,}000$, $\alpha = 0.10$.

b) Results:

Metric	Value
Total seeds	100
Seeds with 0 false positives	93/100
Seeds with 1 false positive	7/100
Seeds with ≥ 2 false positives	0/100
Mean false positive attributions	0.07
Mean per-seed FDR	0.070
FDR $\leq \alpha = 0.10$	YES

Mean per-seed FDR $= 0.070 \leq \alpha = 0.10$, confirming that the BH procedure controls FDR under null conditions over 100 seeds. The 7 false positives (7%) fall within the expected false discovery rate $\alpha = 0.10$ of the BH procedure and are explained by stochastic variation.

Appendix G
Robustness to Inter-Item Correlation

In real data, strong co-occurrence correlations (e.g., beer and snacks) exist, which may violate the independent occurrence assumption of Claim 1 (validity of deduplication). We performed 20-seed evaluation under both Zipf distribution and inter-item correlation conditions.

a) Settings: zipf_only: Zipf-1.0 distribution (independent occurrence). zipf_corr: Zipf-1.0 + correlation probability 0.5–0.7 assigned to 5 item pairs. $N = 100{,}000$, $W = 1{,}000$, $\theta = 100$, 20 seeds.

b) Results:

Condition	Prec	Rec	F1	95% CI	FAR
zipf_only	0.74	0.92	0.81	[0.70, 0.92]	0.25
zipf_corr	0.41	0.85	0.55	[0.45, 0.65]	0.20

F1 = 0.81 for zipf_only, dropping to F1 = 0.55 for zipf_corr. The primary cause of accuracy degradation is that additional dense patterns generated by correlated item pairs increase false positive attributions. This supports the possibility that the independent occurrence assumption of Claim 1 is relaxed in real data; FAR = 0.20 under strong inter-item correlation remains acceptable.

Appendix H
Cross-Signal Effects in OVERLAP Condition

We explain why F1 variance is high (CI width = 0.44) in the OVERLAP condition (§V-B). In the overlap interval $[24{,}000, 28{,}000]$ where E1 $= [16{,}000, 28{,}000]$ and E2 $= [24{,}000, 36{,}000]$ overlap, the item sets of E1 and E2 are simultaneously boosted within the same window. With $\beta = 0.3$, the expected support of cross-patterns (combinations of E1 and E2 items) is $0.3 \times 0.3 \times W = 90 \approx \theta$, and statistical variation causes short dense intervals that exceed θ. Deduplication (Step 5) eliminates most of these by merging them with strong planted patterns, but in some seeds cross-patterns remain as false positives, causing variance in accuracy.

Appendix I
Parameter Selection Rationale

TABLE V: Default Parameters

Parameter	Value
Window size W	1,000
Min. support θ	100
Attribution target patterns	$\lvert P\rvert \geq 2$
Max. pattern length L	unconstrained[1]
Change point detection	threshold crossing
σ	W (= 1,000)
Permutations B	5,000
Significance level α	0.10
Correction method	BH
Magnitude normalization	square root (Eq. (7))
Deduplication	enabled

Rationale for key parameter choices:

- Window size $W = 1{,}000$: Corresponds to 1% temporal resolution for $N = 100{,}000$ transactions. Too small a W makes support estimation unstable; too large reduces temporal precision of change point detection.
- Min. support $\theta = 100$: Sufficiently large relative to the expected baseline support $W \cdot p_{\text{base}}^2 = 1{,}000 \times 0.03^2 = 0.9$, excluding chance co-occurrence while capturing the support increase during boost. $\theta = W/10 = 100$ suppresses the number of false positive patterns under Zipf distribution.
- Decay parameter $\sigma = W$: Proximity decays to $e^{-1} \approx 0.37$ at a distance of one window width. Attribution score decreases rapidly when the change point-to-event boundary distance exceeds W.
- Permutations $B = 5{,}000$: p-value resolution $1/(B + 1) \approx 0.0002$. Guarantees sufficient p-value precision even when testing hundreds of hypotheses simultaneously at $\alpha = 0.10$ with BH correction.
- Significance level $\alpha = 0.10$: Standard FDR control level. Compared to $\alpha = 0.05$, this setting prioritizes detection power, tolerating a false discovery rate below 10%.

[1]Due to Apriori's anti-monotonicity, the search naturally terminates when no patterns satisfying the support threshold exist.

- Magnitude normalization: Square root normalization (Eq. (7)) is adopted. Absolute magnitude has a bias where patterns with higher baseline inflate scores; full normalization ($\div s^{-}$) over-amplifies scores of low-baseline patterns. Square root normalization lies between these extremes and achieves stable accuracy for real-world data with Zipf-like frequency distributions.